PHASE 4 – PROJECT PLANNING

AI FAQ ASSISTANT API

AI-Powered FAQ and Customer Support Automation

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1. INTRODUCTION

Project Planning defines how the **AI FAQ Assistant API** will be developed and completed after the brainstorming, requirement analysis, and system design phases.

The planning phase divides the project into manageable activities and assigns responsibilities to the team members. It also defines the development sequence, major milestones, required resources, testing plan, and expected project outputs.

The project is planned as a RESTful backend application using **Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, and Google Gemini AI**.

The development will be carried out in stages so that each major module can be developed and tested before final integration.

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2. PROJECT DEVELOPMENT PLAN

The project development is divided into the following major stages:

Phase	Development Activity
1	Brainstorming and Ideation
2	Requirement Analysis
3	Project Design
4	Project Planning
5	Project Development
6	Project Testing
7	Project Documentation
8	Project Demonstration

Phase 4 focuses on planning the work that will be carried out during **Project Development, Testing, Documentation, and Demonstration**.

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3. PROJECT MODULE PLAN

The system is divided into major development modules.

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1	Authentication	Registration, login, JWT and profile
2	Database	MongoDB connection and Mongoose models
3	FAQ Management	Create, read, update and delete FAQs
4	FAQ Search	Search FAQ records
5	AI Assistant	AI answers using Gemini
6	AI FAQ Generation	Generate FAQ question and answer
7	Security	Authentication, validation and error handling
8	Testing	API and functional testing
9	Documentation	Screenshots, results and final report

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4. TASK BREAKDOWN

The project is organized into **5 major epics** and **16 tasks/stories**.

Epic 1 – Project Architecture and Setup

Task	Activity
1	Define project architecture
2	Create project structure
3	Configure Node.js and Express.js

Epic 2 – Authentication and Security

Task	Activity
4	Implement user registration
5	Implement user login and JWT
6	Implement authentication middleware

Epic 3 – Database and FAQ Management

Task	Activity
7	Configure MongoDB
8	Create User and FAQ models
9	Implement FAQ CRUD operations
10	Implement FAQ ownership validation

Epic 4 – Search and AI Integration

Task	Activity
11	Implement FAQ search
12	Integrate Google Gemini
13	Implement AI answer generation
14	Implement AI FAQ generation

Epic 5 – Testing and Finalization

Task	Activity
15	Perform API testing
16	Final integration and documentation

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5. TEAM RESPONSIBILITY PLAN

The project team consists of five members.

Team Member	Role	Planned Responsibility
BALRAJ I	Team Lead	Project coordination, architecture, development and integration
BALAKESAVAN S	Team Member	Backend and database activities
BALAMURUGAN E	Team Member	FAQ module and API activities
BHARATH H	Team Member	AI integration and testing support
BHUVANESWARAN G	Team Member	Testing, documentation and project support

Team members may collaborate on different activities depending on the development requirements.

The team lead is responsible for coordinating the overall project and ensuring that the individual modules are integrated into the final system.

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6. DEVELOPMENT SEQUENCE

The development will follow the sequence below:

Step 1 – Project Setup

Create the Node.js project and configure Express.js.

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Step 2 – Database Configuration

Connect MongoDB and configure Mongoose.

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Step 3 – Authentication

Implement registration, login, password hashing, JWT generation, and authentication middleware.

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Step 4 – FAQ Management

Develop FAQ creation, retrieval, updating, deletion, and ownership verification.

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Step 5 – Search

Implement FAQ keyword search functionality.

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Step 6 – AI Integration

Connect Google Gemini and implement AI answer and FAQ generation services.

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Step 7 – Error Handling and Validation

Implement centralized error handling and request validation.

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Step 8 – Testing

Test all major API operations using Postman and the available API testing scripts.

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Step 9 – Final Integration

Integrate all modules and verify the complete system.

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7. PROJECT TIMELINE

The project activities are planned in the following order.

Week	Planned Activity
Week 1	Brainstorming and project idea
Week 2	Requirement analysis
Week 3	System design and architecture
Week 4	Project planning and task allocation
Week 5	Backend and project setup
Week 6	Database and authentication
Week 7	FAQ management
Week 8	FAQ search

Week	Planned Activity
Week 9	Google Gemini integration
Week 10	API testing
Week 11	Bug fixing and final integration
Week 12	Documentation and demonstration preparation

The timeline can be adjusted according to development progress and project requirements.

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8. RESOURCES REQUIRED

Software Resources

Resource	Purpose
Visual Studio Code	Application development
Node.js	Backend runtime
npm	Package management
Express.js	REST API
MongoDB	Database
Mongoose	Database modelling
Postman	API testing
Git/GitHub	Version control and project submission
Google Gemini API	AI functionality

Hardware Resources

Resource	Requirement
Processor	Intel Core i5 or equivalent
RAM	Minimum 8 GB
Storage	Minimum 1 GB free project space

Resource	Requirement
Internet	Required for AI API
Keyboard/Mouse	Required

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9. RISK AND MITIGATION PLAN

During development, several technical risks may occur.

Risk	Possible Effect	Mitigation
MongoDB connection failure	Database operations fail	Verify connection configuration
Invalid JWT	Protected APIs cannot be accessed	Validate token generation and middleware
Gemini API failure	AI functions unavailable	Implement controlled error handling
Invalid user input	Incorrect data	Use request validation
API errors	Application operations fail	Use centralized error handling
Integration issues	Modules may not work together	Perform testing after each module
Dependency issues	Application may not start	Maintain correct package versions
Testing failures	Incorrect functionality	Repeat testing and fix defects

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10. PROJECT MILESTONES AND EXPECTED OUTPUT

Milestone 1 – Initial Planning

Output:

- Project idea
- Problem identification
- Objectives
- Team formation

Milestone 2 – Requirements

Output:

- Functional requirements
- Non-functional requirements
- System requirements

Milestone 3 – Design

Output:

- System architecture
- Database design
- API design
- AI processing flow

Milestone 4 – Backend Development

Output:

- Express server
- Database connection
- Authentication
- FAQ APIs

Milestone 5 – AI Integration

Output:

- Gemini service
- AI answer generation
- AI FAQ generation

Milestone 6 – Testing

Output:

- API test results
- Error handling verification
- Authentication testing
- FAQ testing

- AI testing

Milestone 7 – Finalization

Output:

- Integrated project
- Final source code
- Project documentation
- Demonstration

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11. DOCUMENTATION AND SCREENSHOT PLAN

Screenshots will be collected during the development and testing stages rather than during the planning phase.

Phase	Evidence to Collect
Phase 3	Architecture and design diagrams
Phase 5	VS Code, project structure and implementation
Phase 6	Postman requests and test results
Phase 7	Final implementation screenshots
Phase 8	Final working demonstration

The development screenshots should show the actual project implementation, while testing screenshots should show API requests, responses, and validation results.

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12. FINAL PROJECT PLAN

The complete development plan can be summarized as:

Planning

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Backend Setup

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Database Configuration

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Authentication

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FAQ Management

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Search

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AI Integration

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Testing

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Final Integration

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Documentation

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Demonstration

The planning phase ensures that the project activities are organized into manageable tasks and that each major feature is developed and verified before the final system is completed.

CONCLUSION

The **Project Planning phase** establishes the development roadmap for the AI FAQ Assistant API.

The project has been divided into five major epics covering setup, authentication, database and FAQ management, search and AI integration, and final testing and documentation.

The responsibilities of the team members, development sequence, resources, timeline, risks, milestones, and documentation plan have been identified.

This plan provides a structured path for moving from the project design into **Phase 5 – Project Development**, where the actual implementation of the AI FAQ Assistant API will be carried out.